Many-body expansion of the CT commutators

generated by `expand_commutators.py`

All commutator blocks of the `commutator` stage, taken from its API (`commutator_blocks()` of `commutator/generate_commutators.py`) and expanded with the generalized Wick theorem (`generalized_wick_th`
The truncation is by perturbation order (Møller–Plesset counting $\hat{h} \sim O(1)$, $\hat{V} \sim O(\lambda)$, $\hat{T} \sim O(\lambda)$; the Fock operator differs from $\hat{h}$ only at $O(\lambda)$), correct through $O(\lambda^2)$: the full first-order term $[\hat{H}_b, \hat{A}]$ with $\hat{H}_b = \hat{h} + \hat{V}$, and the Fock half of the second order, $\frac{1}{2}[[\hat{F}, \hat{A}], \hat{A}]$ (writing $\hat{F}$ for $\hat{h}$ there changes the result only at $O(\lambda^3)$). The valence double commutator is $O(\lambda^3)$ and is not expanded. Repeated indices are summed over their space. The index spaces and the delta rules are those of `readme.md`:

$$p, q, r, s: \text{ whole space}, \qquad i, j, k, l: \text{ occupied}, \qquad a, b, c, d: \text{ virtual}, \tag{1}$$

$$\delta(\text{whole}, \text{vir}) \to \text{virtual}, \qquad \delta(\text{whole}, \text{occ}) \to \text{occupied}, \qquad \delta(\text{occ}, \text{vir}) = 0. \tag{2}$$

The operators are

$$\hat{h} = h^q_p \, \hat{E}^p_q, \qquad \hat{V} = \tfrac{1}{2} g^{pq}_{rs} \, \hat{E}^{rs}_{pq}, \tag{3}$$

$$\hat{F} = f^q_p \, \hat{E}^p_q, \qquad \hat{T}_1 = t^a_i \, \hat{E}^a_i, \tag{4}$$

$$\hat{T}_2 = \tfrac{1}{2} t^{ab}_{ij} \, \hat{E}^{ab}_{ij}, \tag{5}$$

with $\hat{h}$, $\hat{F}$ and $\hat{V}$ Hermitian; the second excitation of the nested commutators carries the dummy indices c, d and k, l. The coefficients are real and follow the spin-integrated convention: no index antisymmetry is used anywhere, g^{pq}_{rs} and t^{ab}_{ij} carry only the simultaneous pair swap $g^{pq}_{rs} = g^{qp}_{sr}$, $t^{ab}_{ij} = t^{ba}_{ji}$, and $h^q_p = h^p_q$, $f^q_p = f^p_q$, $g^{pq}_{rs} = g^{rs}_{pq}$. Conjugate blocks are expanded by conjugating the first-order result term by term ($\hat{E}^{u\cdots}_{l\cdots} \to \hat{E}^{l\cdots}_{u\cdots}$, $f^q_p \to f^p_q$, $g^{pq}_{rs} \to g^{rs}_{pq}$; t is real and unchanged). Blocks containing operators of rank higher than two are marked; their cumulant expansion (and the $[\ldots]_{1,2}$ truncation) is done separately. The expanded blocks are available programmatically through `expanded_blocks()`.

Equal terms are merged (`readme.md`, rule 4) over every identity that is exact in this convention: renaming of any summed index within its space — the free indices of the operator are the summation labels of the operator basis sum, so they can be renamed like tensor dummies — the pair symmetries of g and t above, and the diagonal pair-permutation symmetry of the spin-summed $\hat{E}$: the two index rows may only be permuted together, with no sign ($\hat{E}^{qp\cdots}_{rs\cdots} = \hat{E}^{pq\cdots}_{sr\cdots}$); a single row alone is *not* a symmetry of the spin-summed operator.

Every block is verified numerically against the direct nested commutator on Jordan–Wigner matrices, twice: with spin-orbital operators (4 modes, 2 occupied + 2 virtual) and with the genuinely spin-summed operators (4 spatial orbitals = 8 spin orbitals); the second check is the one that sees the merging rules.

1 First-order commutators

$$\left[\hat{h},\,\hat{T}_1\right] = -h_p^i t_i^a \hat{E}_p^a + h_p^a t_i^a \hat{E}_i^p \tag{6}$$

$$\left[\hat{h},\,\hat{T}_2\right] = -h_p^j t_{ij}^{ab} \hat{E}_{ip}^{ab} + h_p^b t_{ij}^{ab} \hat{E}_{ij}^{ap} \tag{7}$$

$$\left[\hat{V},\,\hat{T}_1\right] = -g_{qr}^{ip} t_i^a \hat{E}_{qr}^{ap} + g_{pq}^{ar} t_i^a \hat{E}_{ir}^{pq} \tag{8}$$

$$\left[\hat{V},\,\hat{T}_2\right] = -\tfrac{1}{2}\, g_{pq}^{ij} t_{ij}^{ab} \hat{E}_{pq}^{ab} + \tfrac{1}{2}\, g_{pq}^{ab} t_{ij}^{ab} \hat{E}_{ij}^{pq} \tag{9}$$

$$- g_{qr}^{jp} t_{ij}^{ab} \hat{E}_{iqr}^{abp} + g_{pq}^{br} t_{ij}^{ab} \hat{E}_{ijr}^{apq} \quad \text{(three-body terms)} \tag{10}$$

2 Second order: Fock blocks

$$\left[\left[\hat{F},\,\hat{T}_1\right],\,\hat{T}_1\right] = -2\, f_j^b t_i^b t_j^a \hat{E}_i^a \tag{11}$$

$$\left[\left[\hat{F},\,\hat{T}_1\right],\,\hat{T}_2\right] = -f_k^c t_k^a t_{ji}^{bc} \hat{E}_{ij}^{ab} - f_k^c t_j^c t_{ik}^{ab} \hat{E}_{ij}^{ab} \tag{12}$$

$$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_1\right] = -f_k^c t_{ik}^{ab} t_j^c \hat{E}_{ij}^{ab} - f_k^c t_{ji}^{bc} t_k^a \hat{E}_{ij}^{ab} \tag{13}$$

$$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_2\right] = -2\, f_l^d t_{kj}^{cd} t_{il}^{ab} \hat{E}_{ijk}^{abc} \quad \text{(three-body terms)} \tag{14}$$

$$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_1\right] = f_j^i t_j^b t_i^a \hat{E}_b^a - f_p^b t_i^b t_i^a \hat{E}_p^a \tag{15}$$

$$+ f_b^a t_i^a t_j^b \hat{E}_j^i - f_p^j t_j^a t_i^a \hat{E}_i^p \tag{16}$$

$$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_2\right] = f_k^j t_k^c t_{ji}^{ab} \hat{E}_{ci}^{ab} \tag{17}$$

$$- f_p^c t_j^c t_{ij}^{ab} \hat{E}_{ip}^{ab} + f_c^b t_i^c t_{jk}^{ab} \hat{E}_{jk}^{ai} - f_p^k t_k^b t_{ij}^{ab} \hat{E}_{ij}^{ap} \tag{18}$$

$$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_1\right] = f_k^j t_{ki}^{bc} t_j^a \hat{E}_{bc}^{ai} \tag{19}$$

$$- f_p^c t_{ji}^{bc} t_j^a \hat{E}_{bp}^{ai} - f_p^c t_{ij}^{bc} t_j^a \hat{E}_{pb}^{ai} + f_p^i t_{ji}^{bc} t_j^a \hat{E}_{bc}^{ap} \tag{20}$$

$$+ f_c^b t_{ij}^{ab} t_k^c \hat{E}_{ak}^{ij} + f_p^a t_{ji}^{ab} t_k^b \hat{E}_{kp}^{ij} - f_p^k t_{ik}^{ab} t_j^b \hat{E}_{aj}^{ip} \tag{21}$$

$$- f_p^k t_{ki}^{ab} t_j^b \hat{E}_{ja}^{ip} \tag{22}$$

$$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_2\right] = f^i_j t^{cd}_{jk} t^{ab}_{ik} \hat{E}^{ab}_{cd} \tag{23}$$

$$- f^d_p t^{cd}_{ij} t^{ab}_{ij} \hat{E}^{ab}_{cp} + f^a_b t^{ac}_{ij} t^{bc}_{kl} \hat{E}^{ij}_{kl} - f^l_p t^{ab}_{il} t^{ab}_{jk} \hat{E}^{ip}_{jk} \tag{24}$$

$$+ f^k_l t^{cd}_{li} t^{ab}_{kj} \hat{E}^{abi}_{cjd} - f^d_p t^{cd}_{ki} t^{ab}_{kj} \hat{E}^{abi}_{cjp} \tag{25}$$

$$- f^d_p t^{cd}_{ik} t^{ab}_{jk} \hat{E}^{abi}_{jpc} + f^j_p t^{cd}_{kj} t^{ab}_{ki} \hat{E}^{abp}_{cid} \tag{26}$$

$$+ f^c_d t^{bd}_{ij} t^{ac}_{kl} \hat{E}^{aij}_{kbl} + f^b_p t^{bc}_{ji} t^{ac}_{kl} \hat{E}^{aij}_{klp} \tag{27}$$

$$- f^l_p t^{bc}_{il} t^{ac}_{jk} \hat{E}^{aip}_{jbk} - f^l_p t^{bc}_{li} t^{ac}_{jk} \hat{E}^{aip}_{jkb} \quad \text{(three-body terms)} \tag{28}$$

3 Assembly

With the blocks above, the transcorrelated Hamiltonian through $O(\lambda^2)$ is

$$\left[\hat{H}_b,\,\hat{A}\right] = \left[\hat{h},\,\hat{A}\right] + \left[\hat{V},\,\hat{A}\right], \tag{29}$$

$$\left[\hat{h},\,\hat{A}\right] = \left[\hat{h},\,\hat{T}_1\right] + \left[\hat{h},\,\hat{T}_2\right] + \text{h.c.}, \tag{30}$$

$$\left[\hat{V},\,\hat{A}\right] = \left[\hat{V},\,\hat{T}_1\right] + \left[\hat{V},\,\hat{T}_2\right] + \text{h.c.}, \tag{31}$$

$$\tfrac{1}{2}\left[\left[\hat{F},\,\hat{A}\right],\,\hat{A}\right] = \tfrac{1}{2}\sum_{r,s=1}^{2}\left(\left[\left[\hat{F},\,\hat{T}_r\right],\,\hat{T}_s\right] + \left[\left[\hat{F},\,\hat{T}_r\right]^\dagger,\,\hat{T}_s\right]\right) + \text{h.c.}, \tag{32}$$

the valence double commutator $\frac{1}{2}[[\hat{V},\hat{A}],\hat{A}] \sim O(\lambda^3)$ being dropped by the perturbation-order truncation.